



Fig. 8: Central management in SDN (Courtesy: www.ieit.cnr.it)

with business application plane.

Southbound. Southbound interface allows network components to communicate with lower-level components of that network. In SDN, OpenFlow protocol behaves as southbound, enabling intercommunication of controller and data planes. Extensible Markup Language (XML) and Lisp are other examples of southbound SDN.

Implementation considerations

Three basic aspects that are taken into prime consideration while implementing the new architecture are:

Quality of Service (QoS). Considering the current trend of growing Internet users, the prime factor for acceptance of any new technology is the QoS it provides. The traffic-handling capacity and error rates are the parameters that judge the efficiency of any technology. Protocols for dynamic networking are needed to be designed such that there is no compromise with the QoS it offers.

Security. SDN security requirements differ from those of a static network due to their inherent characteristics. Protocols are expected to reduce the security threats by deploying SDN controllers within their secure computing environments. Being dynamic, SDN is more prone to hazardous security attacks. However, the networking environment needs to be scalable, secured and efficient, simultaneously.

Flexibility. Adaptation of new technology is possible only if its implementation has flexibility. The objective behind switching to a new technology can be achieved by making the technology user-friendly.

Also, scalability is a mandatory characteristic of any network, especially when it is a dynamic architecture as more number of private networks join the existing network every day (the IoT).

Benefits of SDN

Better traffic-handling capacity.

In today's networks, proprietary firmware on the switch determines where frames are forwarded. In SDN, network administrators can actually shape network traffic. They can do this from a centralised network console that integrates the information and controls all their network switches into a kind of network fabric. They can also change data traffic rules on-the-fly if they need to.

The network administrator has complete control over network traffic through a software interface that SDN provides. This allows organisations to decrease their reliance on more expensive switches with proprietary firmware that perform these functions—and that must be set manually.

Relaxing vendors from the burden to embedding specific functionalities into individual devices. Since vendors do not need to embed any single function/program into individual devices,

their task becomes easier and time saving. A common prototype of a device can be approved, and all devices must be manufactured such that these are flexible and programmable.

Central management. SDN provides network administrators an opportunity to make their network device adjustments through a software interface instead of manually configuring individual hardware. Fig. 8 shows the comparison between traditional networking and SDN. It is clearly displayed that, in SDN, no separate network OS is available with individual devices. Rather, a central OS controls each and every connected device, and all features are generalised and not specific for individual device.

To summarise, owing to the increase in the use of the Internet and evolution of new technologies like Cloud computing and network virtualisation, there is need of a new architecture that can offer flexible and affordable services.

SDN is a promising architecture that removes manual configuration and hardware dependence. It is totally software-centric, and the authority to configure and implement different protocols and handle traffic is provided exclusively to the administrator. This promotes efficient use of bandwidth and better handling of traffic. Networks would no longer be a mere collection of various hardware devices. **EFY**